What if `session.user.name` contains something that should never appear in an SQL query? Correct handling of SQL connections, databases, and queries is complicated. When you create a DSL, be careful that what you create is appropriate to the *domain*.

Writing internal DSLs not only means bending the language to fit your needs but also means writing programs that target the needs of users, instead of focusing on implementation details that users don't care about. What could be better than a program that's easy to write? A program that you don't have to write at all! How do you achieve this? Instead of writing programs all the time, you write programs that write other programs. There's one program you already use that does this for you—it's called the CoffeeScript compiler.

8.3. How the compiler works

The CoffeeScript compiler is written in CoffeeScript. The first CoffeeScript compiler was written in Ruby, but since version 0.5 the compiler has been implemented in CoffeeScript. That's right, CoffeeScript is compiled using CoffeeScript, something you should spend some time to ponder, as Scruffy does in figure 8.1. The diagram that Scruffy is holding in figure 8.1 is called a Tombstone Diagram (or T-Diagram). If you're interested in exploring compilers outside of this chapter, then you'll likely come across more of them.

Figure 8.1. The CoffeeScript compiler is written in CoffeeScript.